

Final Project

Total: 130 points

Due Date: May 13th, 2026

Submit all your answers in one notebook file as (final_yourname.ipynb)

All the data files are available in Module 9: Final Project

Question 1 (40 pts)

Sentiment Analysis has become a fundamental task in Natural Language Processing (NLP) with applications in social media analytics, recommender systems, business intelligence, political forecasting, healthcare, and human-computer interaction.

The IMDb dataset (**movie_data.csv**) contains 25,000 highly polarized movie reviews labeled as:

- Positive review = 1
- Negative review = 0

Answer the following:

- a) Read the csv file in python Pandas DataFrame and make **70% as training and 30% as test sets**.
- b) Perform the necessary pre-text processing (e.g., convert to lower case, remove numbers and punctuations, perform stemming and/or lemmatization and remove stop words) for vocabulary generation.
- c) Represent the text (TF-IDF feature weighting) based on two different vocabulary generation using only **unigrams** or **unigrams and bigrams**.
- d) Perform exploratory analysis:
 - Review length distribution
 - Word frequency analysis
 - Positive vs negative review vocabulary difference
 - Show the **IDF** scores only of the first 50 keywords in your vocabulary.

You may visualize using: Histograms/Boxplots/Word clouds

e) Train the dataset using **Logistic Regression, Random Forest and ANN** (at least two hidden layers) classifiers.

f) **Perform K-NN classification:** Fit and test the model for different values for K (from 1 to 25) **using a for loop** and record and plot the KNN's testing accuracy in a variable (scores).

Compare:

- Euclidean distance
- Cosine similarity

Plot: Validation accuracy vs K

g) Compare the performances of different classifiers (test set) based on different feature representations:

Model Evaluation and Statistical Analysis: Classification Metrics, Accuracy, Precision, Recall, F1-score and ROC-AUC

Visualizations: Confusion matrices and ROC curves

Question 2 (60 pts)

The goal is to build and evaluate multiple machine learning pipelines for multi-class emotion classification and semantic clustering using the **Emotions dataset (emotions.zip)**. The dataset contains text documents labeled with emotional categories such as **sadness, anger, joy, fear, love, and surprise**. Dataset is split into train, test & validation for building the machine learning model.

Below are two examples:

*i feel like I am still looking at a blank canvas blank pieces of paper; **sadness***

*i felt anger when at the end of a telephone call; **anger***

Perform the following:

a) Load the train, test and validation datasets from csv files to pandas dataframe.

b) Display: dataset dimensions, class distributions, missing values, and duplicate records. Perform exploratory analysis:

- average sentence length,
- most frequent words,
- class imbalance visualization,
- word clouds for each emotion category.

- c) Perform the necessary pre-text processing (similar to Question 1) for vocabulary generation.
- d) Generate a vocabulary using the **top 500 most frequent terms** from the training corpus only.
- e) Construct multiple feature representations.

1. Bag-of-Words (BoW)

2. TF-IDF

3. Word Embedding Representation (Use pre-trained embeddings):
Word2Vec Or GloVe

Represent each document using average word embeddings

- f) Train the dataset using **Decision Tree, Random Forest, and ANN.**

Hyperparameter Tuning: Use: GridSearchCV Or RandomizedSearchCV for at least two classifiers.

- g) Evaluate all models on the test dataset. Required metrics
 - Accuracy, Precision, Recall, Macro F1-score
 - Confusion Matrix

Create at least **five new sentences** not included in the dataset.

Example:

“I feel completely devastated after losing my wallet.”

“I am excited about my graduation ceremony.”

Predict emotions using the best-performing classifier.

- h) Apply dimensionality reduction techniques on TF-IDF or embedding vectors using **PCA**. Visualize emotion clusters in 2D space.
- i) Perform K-Means clustering on the training dataset.

Experiment with cluster sizes: **K** = 3 to 10

Use:

- Elbow Method
- Silhouette Score

Determine optimal K.

Show the cluster membership of the sentences in the test dataset.

Question 3 (30 pts)

The dataset (image_caption.txt) contains **image IDs** and medical image **captions** (1000) extracted from journal articles.

The objective is to build a multi-stage semantic retrieval system capable of retrieving medically relevant image captions using:

- TF-IDF + Cosine Similarity
 - Static Word Embeddings (Word2Vec/GloVe)
- a) Read the data file in python Pandas DataFrame.
 - b) Perform the necessary pre-processing task (e.g., punctuation, numbers, stop word removal, etc.)
 - c) Create Term-Document Matrix with **TF-IDF weighting**
 - d) Based on the following two queries:
Q1: “angiographic image shows normal coronary artery”
Q2: “CT scan demonstrating pulmonary embolism”

search and display the 10 top ranked image ids (with associated captions) by using **Cosine Similarity measure** to find the similarity between query and documents.

- e) Implement semantic similarity retrieval using: **Word2Vec** or **GloVe**
Compute similarity between **query** embeddings and caption embeddings

Compare: TF-IDF retrieval vs Word embedding retrieval

- f) Visualize caption embeddings using PCA
Color points by:
 - semantic clusters
 - disease types
 - imaging modality

Bonus/ Extra Credit (20 Pts)

Implement a **simple web interface** (using either Flask or Streamlit) to submit your query (as text input) and display the top K (5 or 10) search results for the above queries and create two new queries (related to the dataset) of your own.

You Submit a few screenshots of your web-based search result in a sperate pdf file in addition to the final notebook file.